EXPERIMENTAL PLANNING & PATENT VALIDATION DOCUMENT

AI-ENABLED SMART DIGIKHET PROTECTOR SYSTEM

for Intelligent Crop Safety Using Vision-Based Detection and Autonomous Deterrence Mechanism

Pre-Implementation Field Experiment Plan

Prepared by

Pravash Singh

Founder, Smart Bijuka

Contact: 7247501350 | Email: pravashs775@gmail.com

Document prepared for review by **Project Mentor, Incubation Center Experts**, Patent Consultants and Research Supervisors prior to field implementation.

Status: *Pre-Implementation Planning Stage*

1. Introduction

This document is an Experimental Planning and Patent Validation Document for the AI-Enabled Smart DigiKhet Protector System, also referred to as Smart Bijuka. It is prepared for review before field implementation begins.

This document is not a research paper, not a patent application, and not a project report. It is a planning document. Its purpose is to lay out the experiment clearly enough that a mentor, incubation expert, or patent consultant can review the approach, raise concerns, and approve the plan before any prototype work starts in the field.

The system under review combines computer vision, an embedded controller, physical and audio deterrents, IoT alerts, and solar power to protect crops from animal intrusion without harming the animal. The experiment described here is designed to test that claim directly, under controlled field conditions, before the system is scaled or deployed commercially.

2. Project Background

Crop damage from animal intrusion is a recurring problem for farmers in rural and semi-remote areas. The usual responses are static scarecrows, manual night guarding, electric fencing, or basic motion alarms. Each of these has a clear weakness:

- Scarecrows lose effectiveness once animals get used to them.
- Manual guarding is exhausting, unsafe at night, and not scalable.
- Electric fencing is expensive to install and risky to maintain.
- Motion sensors and basic CCTV cannot tell the difference between a human and an animal, so they generate false alerts.

The Smart DigiKhet Protector system, branded as Smart Bijuka, is built to close these gaps. It uses a camera and a YOLOv8-based detection model to classify what enters the field. An ESP32 controller then decides what to do: stay silent for humans and activate a servo-driven deterrent with directional audio for animals. The system runs on solar power with a battery backup, so it works without a grid connection, and it sends real-time alerts to the farmer's phone through an IoT module.

The patent filed for this system describes one independent claim covering the full system architecture, and nine dependent claims covering specific components and behaviours: the YOLOv8 model, the ESP32 controller, the servo actuation, the selective audio logic, the IoT notifications, the solar power unit, the cooldown mechanism, the continuous monitoring loop, and the modular design. This experimental plan is built around testing those exact claims in a real field setting.

3. Problem Statement

Before this system can be deployed on a real farm, scaled into a product, or carried into a full patent filing, it needs evidence that it actually works the way the patent describes. Right now, the system exists as a design and a set of claims. It has not been tested in a field with live animal movement.

The core technical question this experiment answers is straightforward: when an animal enters the field and moves toward a crop plot, does the system detect it, identify which Smart Bijuka unit is closest to the threat, activate that unit, and guide the animal back toward a safe path, without touching the animal and without setting off the wrong unit?

A secondary question matters just as much: does the system correctly leave humans alone? A system that scares off farm workers, family members, or visitors is not usable, no matter how well it handles animals.

This document plans the experiment that will answer both questions with field data instead of assumptions.

4. Purpose of Experiment

The experiment exists to do four things:

1. Confirm that the patent claims hold up under real field conditions, not just on paper.
2. Collect concrete data on detection accuracy, response time, and correct unit activation that can be shown to a mentor, an incubation panel, or a patent consultant.
3. Surface any gaps in the design before time and money go into a full prototype build.
4. Build the foundation for a future IEEE submission and a complete patent filing, both of which need experimental evidence, not just a working concept.

5. Experimental Objectives

Specific objectives for this experiment are:

- Verify that the AI model can detect an animal entering the field and classify it correctly against a human.
- Verify that the system identifies which crop plot is under threat and activates only the nearest Smart Bijuka unit.
- Verify that the servo and directional speaker combination redirects the animal back toward the central corridor without physical contact.
- Verify that the animal can reach the Exit Gate without entering a protected zone.
- Verify that the system resets correctly and continues monitoring after each intrusion event, including repeated attempts.
- Collect response time and activation accuracy data across all four crop plots and multiple trial runs.

6. Experimental Field Layout

The experiment will run in a field of approximately 200 square feet, laid out as follows:

- Four crop plots, arranged in a 2x2 grid, each representing a protected agricultural zone.
- One central pathway running from the Entry Gate to the Exit Gate, between the four plots. This is the Safe Navigation Corridor.
- One Entry Gate at one end of the corridor and one Exit Gate at the other end.
- Four Smart Bijuka units, one positioned at the outer corner of each crop plot, each fitted with a camera, ESP32 controller, servo, directional speaker, solar panel, and IoT module.



7. System Architecture Overview

The Smart Bijuka system is built from eight components working together: a camera module, an AI processing unit, an embedded controller, a servo-driven actuation unit, an audio unit, an IoT communication module, a PIR sensor, and a solar power unit. The diagrams below, taken directly from the patent drawings, show how these pieces connect and how a detection event moves through the system.

7.1 Block Diagram of System Architecture

Figure 1 shows the high-level architecture. The camera and PIR sensor form the input side. The ESP32 controller sits in the middle and runs the decision logic, fed by the AI processing unit's classification output. The output side covers four response channels: servo actuators for physical movement, the audio unit for deterrent sound, the IoT module for mobile alerts, and LED indicators for visual status. The farmer's mobile app sits at the far end, receiving alerts from the IoT module.

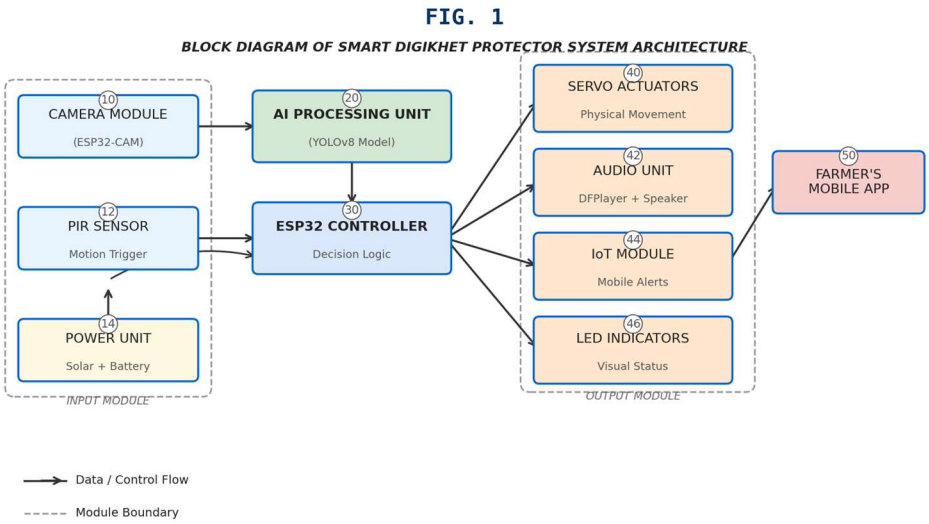


Figure 1. Block Diagram of Smart Bijuka System Architecture

7.2 Flowchart of System Operation and Decision Logic

Figure 2 shows the full operating loop, from system boot to reset. The PIR sensor triggers image capture. The AI model then classifies what was captured into one of three outcomes: no intrusion, human, or animal. Human detections lead to passive monitoring and a silent log entry, no audio, no servo. Animal detections trigger the servo, the audio deterrent, and an IoT alert, followed by a cooldown period before the system resets and resumes monitoring.

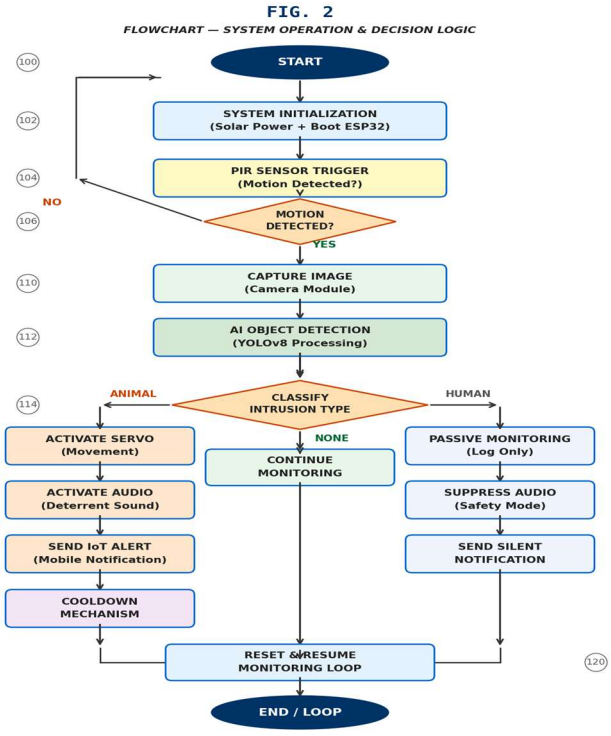


Figure 2. Flowchart of System Operation and Decision Logic

7.3 Physical Layout in the Agricultural Field

Figure 3 shows how a single Smart Bijuka unit is physically built and mounted: a pole-mounted enclosure carrying the camera, ESP32 and AI unit, two servo arms, an audio speaker, a PIR sensor, a solar panel on top, and a battery backup lower on the pole. This same physical design is repeated at all four corners of the experimental field, one unit per crop plot.

FIG. 3

PHYSICAL LAYOUT — SMART DIGIKHET PROTECTOR IN AGRICULTURAL FIELD

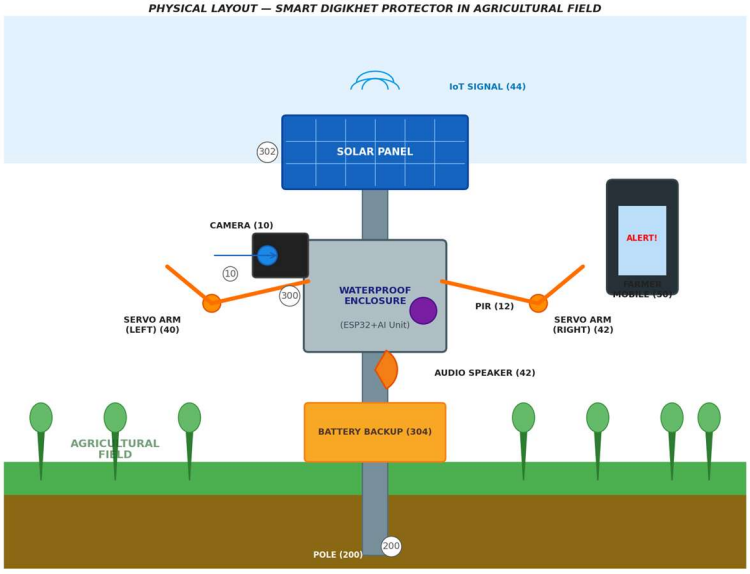


Figure 3. Physical Layout of Smart Bijuka Unit in Agricultural Field

7.4 Component-Level Architecture

Figure 4 breaks each module down to its working parts. This is the level of detail the hardware validation plan in Section 11 will test against.

FIG. 4

COMPONENT-LEVEL ARCHITECTURE — CAMERA, ESP32, ACTUATORS, IoT & SOLAR UNIT

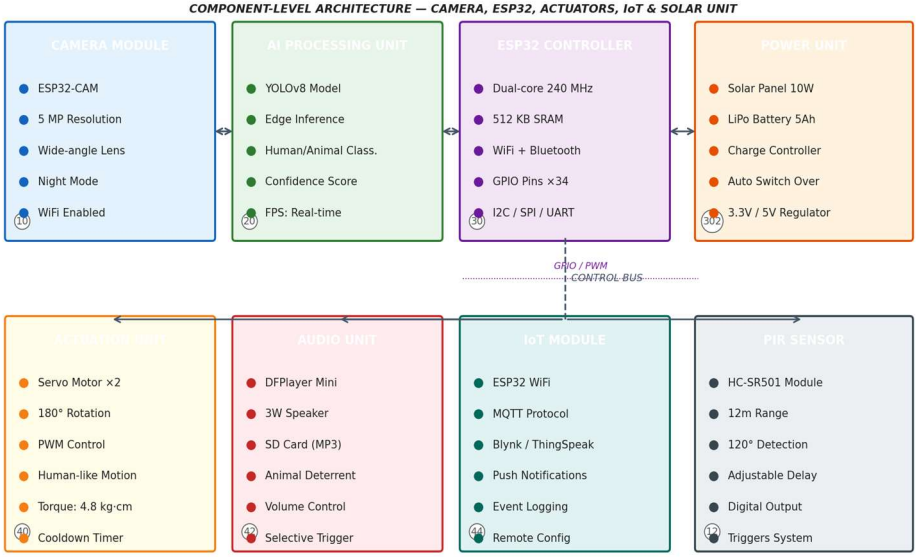


Figure 4. Component-Level Architecture

8. Experimental Workflow

This section lays out the exact sequence of events the experiment is built to observe, from the moment an animal enters the field to the moment it exits. Each step below corresponds to a stage in the flowchart in Figure 2, applied specifically to the four-plot field layout.

Step 1: Animal Entry

The animal enters the field through the Entry Gate and steps onto the central corridor. The PIR sensors on the nearest Smart Bijuka units detect motion and trigger image capture.

Step 2: Continuous Monitoring

The camera feeds continue capturing frames as the animal moves. No deterrent is active at this stage; the system is only watching.

Step 3: AI Detection

The YOLOv8 model processes each captured frame to detect whether an object of interest is present in the scene.

Step 4: Object Classification

The model classifies the detected object as human or animal. If the object is human, the system moves to passive monitoring and takes no further action.

Step 5: Decision Making

The ESP32 controller on the relevant unit evaluates the animal's position relative to the corridor boundary and the nearest crop plot edge.

Step 6: Safe Corridor Check

If the animal's position is still within the corridor, the system takes no action and returns to continuous monitoring.

Step 7: Crop Entry Attempt Detected

If the animal's path crosses toward a crop plot boundary, the system identifies this as an intrusion attempt and proceeds to activate a response.

Step 8: Nearest Smart Bijuka Selection

The controller identifies which of the four units is closest to the point of attempted entry and activates only that unit. No other unit responds.

Step 9: Servo Activation

The selected unit's servo arms move to produce a human-like deterrent motion, signaling presence without making contact with the animal.

Step 10: Directional Speaker Activation

The same unit's directional speaker plays a deterrent sound aimed toward the animal's position, reinforcing the servo movement.

Step 11: Animal Guidance

The combined servo and audio response is expected to redirect the animal's movement away from the crop plot and back toward the central corridor.

Step 12: Return to Corridor

The animal moves back into the safe corridor. The activated unit enters a cooldown period to avoid retriggering on the same animal immediately.

Step 13: Continuous Monitoring Resumes

The system resumes passive monitoring of the corridor as the animal continues moving.

Step 14: Exit

The animal reaches the Exit Gate and leaves the field without having entered any protected crop plot.

If the animal attempts entry into more than one plot during a single pass, steps 7 through 13 repeat independently for each attempt, with only the relevant nearest unit activating each time. This repeated-attempt behavior is tested directly in Scenario 6 in Section 14.

9. Patent Claim Validation

The filed patent contains one independent claim describing the full system, and nine dependent claims describing specific components and behaviors. The purpose of this section is to map each claim to a concrete experimental activity, so that the field test produces direct evidence for or against each claim rather than general impressions of how the system performed.

Claim	Description	Experimental Activity	Expected Behaviour	Validation Method	Evidence to Collect
Claim1 (Independent)	Full system: camera, AI unit, controller, actuator, audio unit, IoT module, and solar power, selectively activating deterrence based on intrusion type.	Run the complete field experiment across all seven scenarios in Section 14, observing the full chain from detection to deterrence to reset.	System detects intrusion, classifies it, and activates only the appropriate response for animals, while staying passive for humans.	End-to-end observation across all scenarios; cross-check against the individual claim validations below.	Full scenario logs, video recordings, response time records, and the consolidated data tables in Section 15.
Claim 2	AI processing unit uses a YOLOv8-based model for classification.	Run the deployed model against test footage and live field passes; verify it is in fact the YOLOv8 architecture in use.	Correct human-versus-animal classification at a rate suitable for reliable field deployment.	Compare classification output against manually verified ground truth for each pass.	To Be Finalized During Experimental Validation.
Claim 3	Embedded controller is an ESP32 microcontroller performing edge-based decision-making.	Confirm the ESP32 processes the classification result and executes the trigger decision on-device, without depending on cloud processing for the decision itself.	Decision logic runs locally with no added latency from network round trips.	Bench test with network disabled to confirm local decision logic still functions; field test to measure decision latency.	To Be Finalized During Experimental Validation.
Claim 4	Actuation unit uses servo motors that generate human-like movements for	Observe servo arm motion during each animal-triggered scenario.	Servo movement is visible, repeatable, and does not make physical contact with the animal.	Visual review of field recordings; confirm no contact occurs across all trials.	Video evidence and a contact/no-contact log for every triggered event.

Claim	Description	Experimental Activity	Expected Behaviour	Validation Method	Evidence to Collect
	animal deterrence.				
Claim 5	Audio unit activates only during animal or mixed intrusion events, not for human-only entry.	Run human-only entry passes and animal-only entry passes separately, then mixed passes, and check audio activation in each case.	Audio stays silent for human-only entry and activates only for animal or mixed intrusion.	Direct observation and audio-trigger log compared against entry type for every test pass.	Trigger log marking entry type against audio activation, pass by pass.
Claim 6	IoT module provides real-time mobile notifications to a remote user.	Trigger an animal detection event and measure the time until the alert appears on the farmer's mobile device.	Notification arrives promptly after the triggering event, with no missed alerts.	Timestamp comparison between event time and notification receipt time, logged for each trial.	Notification delivery log with timestamps for every animal-triggered event.
Claim 7	Solar-powered unit includes a solar panel and rechargeable battery enabling off-grid operation.	Run the system on solar and battery power only, with no external power input, across the full test period.	System remains operational throughout testing without external power, including during low-sunlight periods if battery backup is sufficient.	To Be Finalized During Experimental Validation.	To Be Finalized During Experimental Validation.
Claim 8	Cooldown mechanism prevents repeated triggering of deterrent actions.	Run Scenario 6 (repeated intrusion attempts, Section 14) directly against this claim.	After a deterrence event, the same unit does not retrigger immediately if the animal lingers nearby; it waits out a cooldown period before it can trigger again.	Timestamp the gap between consecutive activations of the same unit during repeated approach attempts.	Activation timestamp log for each unit across repeated-attempt trials.
Claim 9	System operates in a continuous monitoring loop with autonomous reset functionality.	Observe the system across the full duration of testing, including idle periods with no animal present.	System remains in an active monitoring state between events without manual restart, and resets cleanly after each triggered event.	Continuous observation log noting any stalls, freezes, or required manual intervention.	Uptime and reset log for the full testing period.
Claim 10	System is modular and scalable to integrate additional sensors or upgraded AI models.	Not directly testable through behavioral field trials; this claim concerns design architecture rather than field performance.	Component-level documentation shows independent, swappable modules (camera, AI model, controller, actuator, audio, IoT, power) rather than a fixed monolithic build.	Architecture review against Figure 1 and Figure 4, confirming modules connect through defined interfaces rather than hard-coded dependencies.	Component architecture diagram and a written note on interface points between modules.

10. Experimental Scenarios

Seven scenarios are planned to cover the full range of behaviour the system needs to handle correctly. The first five test the core path: a clean pass through the corridor, and an attempted entry into each of the four crop plots. The sixth tests repeated pressure on the same unit. The seventh confirms the animal can complete its journey and leave the field. Each scenario is described with its starting condition, what the AI is expected to decide, which unit should respond, what the system should do, and what counts as the right outcome.

Scenario 1: Animal Walks Correctly Through the Corridor

Field	Detail
Initial Condition	Animal enters through the Entry Gate and moves along the central pathway, staying within the corridor boundary.
AI Decision	Classified as animal; position confirmed to remain inside the safe corridor at all times.
Activated Smart Bijuka	None. No unit activates because no plot boundary is approached.
Expected System Behaviour	All four units remain in passive monitoring. No servo movement, no audio, no IoT alert.
Expected Outcome	Animal proceeds toward the Exit Gate undisturbed. This is the baseline case confirming the system does not over-trigger on normal corridor movement.

Scenario 2: Animal Attempts Crop Plot 1

Field	Detail
Initial Condition	Animal moves off the corridor toward Crop Plot 1's boundary.
AI Decision	Classified as animal; position crosses the threshold near Plot 1.
Activated Smart Bijuka	Smart Bijuka unit assigned to Plot 1, and only that unit.
Expected System Behaviour	Unit 1 activates servo movement and directional audio aimed at the animal's position. Units 2, 3, and 4 remain inactive.
Expected Outcome	Animal redirects away from Plot 1 and back toward the corridor without physical contact.

Scenario 3: Animal Attempts Crop Plot 2

Field	Detail
Initial Condition	Animal moves off the corridor toward Crop Plot 2's boundary.
AI Decision	Classified as animal; position crosses the threshold near Plot 2.
Activated Smart Bijuka	Smart Bijuka unit assigned to Plot 2, and only that unit.
Expected System Behaviour	Unit 2 activates servo movement and directional audio. Units 1, 3, and 4 remain inactive.
Expected Outcome	Animal redirects away from Plot 2 and back toward the corridor without physical contact.

Scenario 4: Animal Attempts Crop Plot 3

Field	Detail
Initial Condition	Animal moves off the corridor toward Crop Plot 3's boundary.
AI Decision	Classified as animal; position crosses the threshold near Plot 3.
Activated Smart Bijuka	Smart Bijuka unit assigned to Plot 3, and only that unit.
Expected System Behaviour	Unit 3 activates servo movement and directional audio. Units 1, 2, and 4 remain inactive.
Expected Outcome	Animal redirects away from Plot 3 and back toward the corridor without physical contact.

Scenario 5: Animal Attempts Crop Plot 4

Field	Detail
Initial Condition	Animal moves off the corridor toward Crop Plot 4's boundary.
AI Decision	Classified as animal; position crosses the threshold near Plot 4.
Activated Smart Bijuka	Smart Bijuka unit assigned to Plot 4, and only that unit.
Expected System Behaviour	Unit 4 activates servo movement and directional audio. Units 1, 2, and 3 remain inactive.
Expected Outcome	Animal redirects away from Plot 4 and back toward the corridor without physical contact.

Scenario 6: Repeated Intrusion Attempts

Field	Detail
Initial Condition	Animal returns to the same plot boundary multiple times within a short window, after already being redirected once.
AI Decision	Classified as animal on each approach; position again crosses the same plot's threshold.
Activated Smart Bijuka	Same unit as the original attempt, subject to the cooldown mechanism described in Claim 8.
Expected System Behaviour	First approach triggers full response. If the animal returns within the cooldown period, the system's behaviour during that window is To Be Finalized During Experimental Validation. Outside the cooldown window, the unit should be able to trigger again normally.
Expected Outcome	This scenario is the direct field test for the cooldown logic. The expected outcome and exact timing thresholds are To Be Finalized During Experimental Validation. pending the cooldown duration set in the controller logic.

Scenario 7: Animal Successfully Exits

Field	Detail
Initial Condition	Animal has traveled the length of the corridor, with or without prior redirection events, and approaches the Exit Gate.
AI Decision	Classified as animal; position confirmed within the corridor as it nears the exit.
Activated Smart Bijuka	None, assuming the animal is within the corridor at this point and not approaching a plot boundary.
Expected System Behaviour	System remains in passive monitoring as the animal exits. No unit activates at the exit point itself.
Expected Outcome	Animal leaves the field through the Exit Gate with zero crop plots entered during the full pass. This is the overall success condition for the experiment.

11. Success Criteria

Success on this experiment is defined by specific, observable outcomes, not a general sense that the system seemed to work. The list below is what gets checked off after each test run.

- Animal reaches the Exit Gate during every trial run.
- Zero crop plot intrusions occur across all trials in Scenarios 2 through 5.
- The correct Smart Bijuka unit activates for each plot, every time, with no exceptions.
- The wrong Smart Bijuka unit never activates, meaning Units 2, 3, and 4 stay silent when Plot 1 is approached, and so on for each unit.
- Servo motion activates correctly and without any physical contact with the animal.
- Directional audio activates correctly and only during animal or mixed intrusion, never for human-only entry.
- IoT notification reaches the farmer's mobile device for every animal-triggered event.
- The system resumes passive monitoring automatically after each event, with no manual restart needed.
- The system runs continuously across the full testing session without requiring intervention.

12. Conclusion

This document lays out a complete plan for testing the Smart DigiKhet Protector system in a real field, before any prototype work begins. It connects every patent claim to a specific test, every test to a specific expected outcome, and every outcome to a way of recording what actually happened.

Nothing in this plan assumes results that have not yet been collected. Wherever a number, specification, or outcome depends on data that does not exist yet, this document says so directly, marked clearly as To Be Finalized During Experimental Validation. The goal is to walk into the mentor meeting with a plan strong enough to survive scrutiny, not a plan padded with invented numbers.

Once this plan is reviewed and any changes from the discussion points in Section 18 are incorporated, the next step is implementation: building the four-unit field setup as described and running the seven scenarios to begin producing real data against this framework.

